*CSC126: Fundamentals of Algorithms and Computer Problem Solving*

UNIVERSITI TEKNOLOGI MARA PAHANG

# ASSESSMENT #1

**QUIZ – 1 HOUR (10%)**

Student Name: .....

Student ID: ..... Group: .....

Lecturer's Name: .....

CLO1 – Apply basic concepts and algorithms in computer programming.

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**SELF DECLARATION**

1. I know that plagiarism is wrong. Plagiarism is to use other's work and pretend that it is one's own.
2. This quiz is my own work.
3. I have not involved, and will not allow anyone to copy my work with the intention of passing it off as their own work.
4. I acknowledge that copying someone else's work (or part of it) is wrong and declare that my quiz is my own work.

**Answer ALL questions.**

**QUESTION 1**

a. Define an algorithm

(2 marks)

b. Write a C++ statement to declare the following:

i. Two integer variables called num1 and num2 and initialize both to zero.
ii. Declare four variables whereby the first two are float and the last two variables are integer.
iii. Declare a variable named employeeName as character. The length of variable is 30.

(6 marks)

c. Write a C++ statement that is equivalent to expression below:

$$C = \sqrt{\frac{a^2 + b^2 - 2ab}{4ac^2}}$$

(2 marks)

**QUESTION 2**

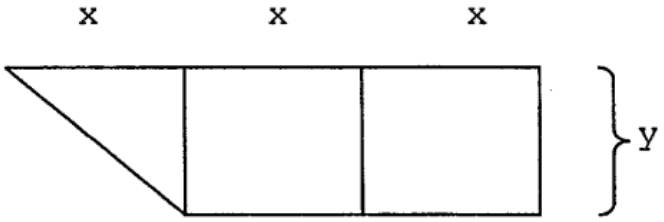
i. Construct a flowchart from the pseudocode below to read and display THREE (3) variables.

```
Start
    Read name, student id, age
    Display name, student id, age
End
```

(5 marks)

ii. Based on the following problem:

You are assigned to write a program to determine the area for the following diagram. Prompt the user to input two integers x and y, then calculate the area and display the value.




i. Write the analysis of this program.

(5 marks)

ii. Draw the flowchart for the above problem.

(5 marks)

iii. Write a C++ program segment for the given problem.

(5 marks)